

Doc 1 — Study Brief

Read tonight. Wireless fundamentals you need in your head before you walk in. Plainsboro Public Library passive site survey, 28 Aug 2026.

1. The three kinds of survey

Type	What it is	When you use it	Are you doing it?
Passive	Phone listens to AP beacons while you walk. Records RSSI, noise, channel, BSSID. Never associates for measurement.	Auditing an existing network. Finding dead zones and interference.	YES — this is tomorrow
Active	Device associates to the SSID and measures real throughput / retries / roaming.	Troubleshooting performance complaints, validating after a fix.	Optional, a few spot checks
Predictive (Planning)	No site visit. Software simulates coverage from a floor plan, wall materials, and virtual AP placement.	Designing a network that doesn't exist yet. Quoting a job remotely.	Later, at your desk, as the 'proposed fix' half of the report

2. Reading dBm — the only scale that matters

RSSI is measured in dBm and is always negative. Closer to zero is stronger. It is logarithmic: every 3 dB is double or half the power, every 10 dB is 10×. That is why –60 is dramatically better than –70, not 'a bit better.'

Signal	Quality	What it means in practice
–30 to –45 dBm	Excellent	You are essentially standing under the AP. Max data rates.
–46 to –60 dBm	Very good	Solid for anything including video calls.
–61 to –67 dBm	Good	–67 is the classic design floor for voice/video. Design to this edge.
–68 to –72 dBm	Marginal	Web and email survive. Rates drop hard. Roaming gets sticky.
–73 to –80 dBm	Poor	Usable only in name. This is what a 'dead zone' complaint actually is.
Below –80 dBm	Unusable	Client will drop or refuse to connect.

Design target for your report: primary coverage areas should be –67 dBm or better on 5 GHz, with SNR above 20 dB. Anything worse is a finding.

3. Noise and SNR

Noise floor is the background RF energy with no WiFi in it — typically around –90 to –95 dBm indoors. **SNR** (signal-to-noise ratio) is simply signal minus noise, expressed in dB. Signal of –65 with noise at –92 gives SNR of 27 dB.

SNR	Verdict
40 dB+	Excellent. Top modulation rates available.
25–40 dB	Very good. Comfortable for voice and video.
15–25 dB	Fair. Data works, rates are reduced.
10–15 dB	Poor. Retransmissions climb, throughput collapses.
Under 10 dB	Effectively unusable.

Why this matters: strong signal with bad SNR is still a broken link. If you find an area with decent RSSI but poor SNR, you have found interference, and that is a much more interesting finding than a plain dead zone.

4. SIR / interference

NetSpot's third heatmap is SIR — signal-to-interference ratio. Interference here means *other WiFi* on the same or overlapping channel, as opposed to plain noise. Two flavors:

- **Co-channel interference (CCI):** two APs on the exact same channel. They can hear each other, so they politely take turns. Not corruption — contention. Everyone gets slower. This is the most common real-world problem in over-deployed networks.
- **Adjacent-channel interference (ACI):** APs on partially overlapping channels (e.g. 2.4 GHz channels 1 and 3). They cannot decode each other, so they transmit over each other. This is genuine corruption and is much worse than CCI.

5. The bands

Band	Channels you should see	Range vs speed	Notes
2.4 GHz	Only 1, 6, 11 are non-overlapping (in the US)	Longest range, slowest, most congested	Every microwave, Bluetooth device, and neighbor lives here. If you see APs on 3, 4, 8, 9 — that is a finding.
5 GHz	36–48 (UNII-1), 52–144 (UNII-2, DFS), 149–165 (UNII-3)	Shorter range, much faster, far cleaner	DFS channels must vacate if radar is detected. Some client devices avoid them.
6 GHz (WiFi 6E/7)	Very wide, very clean	Shortest range, fastest	Only if the library has 6E gear and your phone supports it. Note whether you see any.

Channel width tradeoff: wider channels (40/80/160 MHz) mean more throughput per client but fewer non-overlapping channels available, so more CCI. In a dense public space, narrower is often better. NetSpot shows this as e.g. 'channel: 100 (106)' — primary channel 100, centered at 106, meaning an 80 MHz-wide channel.

6. Worked example — your own apartment scan

From the Inspector screenshots you already took. Practice reading it, because you will do exactly this at the library:

SSID / BSSID	Reading	What it tells you
Verizon_7XFC94 38:06:E6:50:95:66	−26 dBm, 2.4 GHz, ch 11	Extremely strong — you are next to the router. Channel 11 is a correct non-overlapping choice.
Verizon_7XFC94 38:06:E6:50:95:67	−34 dBm, 5 GHz, ch 100 (106)	Same SSID, different BSSID = the 5 GHz radio in the same box. 80 MHz wide, on a DFS channel.
[Hidden SSID] 3E:06:E6:50:95:67	−35 dBm, 5 GHz, ch 100 (106)	Note the BSSID is one bit off from the 5 GHz radio. That is a virtual AP on the same physical radio — almost certainly a guest or IoT network. Hiding the SSID is not security; you found it anyway.
TP-Link_2B20 / FiOS-0MI7X	−51 to −63 dBm	Neighbors. On 2.4 GHz they are your interference. Note the Channels tab rated every 2.4 GHz channel 'Weak' with 3–5 APs each — that band is saturated where you live.

Transferable lesson: one SSID can be many BSSIDs. At the library, the same guest SSID will appear from every AP in the building, each with its own BSSID. Counting distinct BSSIDs is how you count APs. Track BSSIDs, not names.

6b. Your instrument and what it cannot see

You are surveying with a **Motorola Moto G Stylus 5G (2024)**. Know its specification, because every limitation below has to appear in your report's methodology section — and naming them correctly is exactly the kind of thing that reads as competence.

Property	Your device	Consequence for the survey
WiFi standard	802.11 a/b/g/n/ac (Wi-Fi 5), dual-band	You cannot associate at Wi-Fi 6 rates. If the library runs 802.11ax APs, you observe them as ac.
Bands supported	2.4 GHz and 5 GHz only — no 6 GHz	If the library has Wi-Fi 6E/7 access points, the entire 6 GHz band is invisible to you . This is the single biggest blind spot in your survey and must be stated plainly.
Spatial streams	1×1 (your 433 Mbps link on an 80 MHz channel is the giveaway: 80 MHz, one stream, MCS9)	Observed throughput will be roughly half what a 2×2 laptop would achieve on the same AP. Never report your speed test numbers as the network's capability.
Radio calibration	Consumer chipset, uncalibrated, vendor-specific RSSI reporting	Absolute dBm values carry maybe ±3–6 dB of uncertainty versus a professional survey adapter. Relative comparisons across your own survey are still valid — that is what makes the work useful.
Antenna	Internal, single, orientation-dependent	Why device orientation consistency matters so much. Change how you hold it and you change the data.

How to say this in an interview: "I used a single-stream Wi-Fi 5 handset, so my absolute figures carry a few dB of uncertainty and I had no visibility into 6 GHz. I designed the survey around relative comparison within a consistent measurement method, which is where the findings actually come from." That sentence does more for you than any heatmap.

6c. Reading NetSpot's 'map coverage' number correctly

Your practice project showed **6 data points · map coverage 75%, good**. Six points cannot characterize a library floor. So what is the number measuring?

It measures **spatial spread** — how much of the floor plan area falls within interpolation range of at least one point. It says nothing about density or accuracy. Six points spread to the far corners will read 'good' while producing a heatmap that is 90% guesswork between them.

Treat map coverage as necessary but nowhere near sufficient. High coverage with low density is the most misleading output this tool can produce, because it looks authoritative. Your real target is point *density*: a reading every 15–20 ft, and extra density wherever signal changes fast. Report both numbers — point count and coverage % — so a reader can judge for themselves.

7. Attenuation cheat sheet

Approximate signal loss when passing through common materials. Use these to explain *why* a dead zone exists. Values vary; treat as order-of-magnitude.

Obstacle	Approx. loss (5 GHz)	Obstacle	Approx. loss (5 GHz)
Drywall / plasterboard	3–5 dB	Solid wood door	5–10 dB
Interior glass window	3–8 dB	Brick wall	8–15 dB
Concrete wall	15–25 dB	Reinforced concrete / core	20–30 dB+
Loaded bookshelves	5–15 dB	Metal shelving / filing	15–30 dB
Elevator shaft	30 dB+ (effectively opaque)	Human body	3–6 dB

Obstacle	Approx. loss (5 GHz)	Obstacle	Approx. loss (5 GHz)
Low-E / coated glass	15–30 dB	Aquarium / water volume	10–20 dB

A library is an unusually good teaching environment for this: dense paper stacks, glass study rooms, and a concrete core in one building.

8. Glossary — know these cold

Term	Meaning
SSID	The network name humans see. Can be broadcast from many APs at once.
BSSID	MAC address of one radio on one AP. This is the unique identifier you actually track.
RSSI	Received signal strength, in dBm. What the heatmap colors.
Noise floor	Background RF with no WiFi in it. Usually –90 to –95 dBm indoors.
SNR	Signal minus noise, in dB. The real predictor of link quality.
SIR	Signal minus competing WiFi. NetSpot's interference heatmap.
CCI / ACI	Co-channel (same channel, contention) vs adjacent-channel (overlapping, corruption).
DFS	Dynamic Frequency Selection. 5 GHz channels shared with radar; AP must vacate on detection.
Cell / coverage cell	The area one AP radio effectively serves.
Cell overlap	Deliberate coverage overlap between neighboring APs, typically 15–20%, so clients can roam without dropping.
Sticky client	A device clinging to a distant weak AP instead of roaming to a closer strong one.
Band steering	AP nudging dual-band clients onto 5 GHz instead of 2.4 GHz.
Capacity vs coverage	Coverage = can you hear it. Capacity = can it serve everyone at once. A library at 3pm is a capacity problem, not a coverage one.
AP-on-a-Stick (APoS)	Survey method: temporarily mount an AP, measure its real cell, repeat. The physical alternative to predictive.
Data point	One measurement at one map location. Your unit of work.
Map coverage %	NetSpot's estimate of how much of the floor plan your points actually justify. Low = untrustworthy heatmap.

9. What good findings look like

You are not going to the library to make a pretty picture. You are going to produce statements like these. Read them and keep them in mind while you walk:

- "5 GHz coverage falls below –72 dBm throughout the northeast reading area; the nearest AP is roughly 60 ft away with two stack rows and a concrete column in the path."
- "Three APs are broadcasting on 2.4 GHz channel 6 within audible range of one another, producing co-channel contention across the main floor."
- "The guest SSID is readable at –61 dBm from the parking lot, extending the attack surface outside the building envelope."
- "Signal is adequate but SNR drops to 12 dB near the program room, suggesting a non-WiFi noise source in that area."
- "2.4 GHz and 5 GHz share an SSID with no apparent band steering; older clients may associate to 2.4 GHz despite strong 5 GHz availability."

The discipline: every finding = observation + measurement + probable cause. Numbers without a cause is data, not analysis. Cause without numbers is a guess.